

RISC-V

The road to ubiquity

OSHcamp - August 2026



It's 2028, your Flutter app is in the top 10
Huawei's latest foldable flagship has a RISC-V SoC



Hi, I'm Chris

@cpswan

<https://chris.swanz.net>



**dart-docker**

Public

Unwatch

main ▾

14 Branches

0 Tags

Go to file

T

+

<> Code ▾

**Dart CI** Upgrade dart to 3.13.2 ✓

83bee4f · 3 days ago

376 Commits



.github

Bump actions/checkout from 7.0.0 to 7.0.1...

3 weeks ago



beta/trixie

Upgrade dart to 3.13.1

last week



scripts

Bump test from 1.31.1 to 1.31.2 in /scripts ...

last month



stable/trixie

Upgrade dart to 3.13.2

3 days ago



Dockerfile-debian.template

feat: Add RISCv64 to Dockerfile template

last year



LICENSE

Create LICENSE

5 years ago



OWNERS

Add myself to dart docker owners and mai...

last year



README.md

Move usage docs to to the right folder (#15)

5 years ago



versions.json

Upgrade dart to 3.13.2

3 days ago

Agenda

- RISC-V overview
- RISE worked
- RISC-V cloud is (just about) a thing :)
- Hardware vapourware
- The China factor dissipated
- New shoots are sprouting

RISC-V overview



RISC-V is an open standard
Instruction Set Architecture (ISA)
enabling a new era of processor
innovation through open
collaboration

<https://riscv.org/about/>



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Can I run Dart on my soldering iron?

PINECIL – Smart Mini Portable Soldering Iron



PLATFORM	• OS: Ralim's IronOS build • Chipset: Bouffalo BL-706 • CPU: 32-bit RV32IMAFRC RISC-V "SiFive E24 Core" @ 144 MHz
DISPLAY	• Type: OLED White Colour Monochrome Display • Size: 0.69 inches • Resolution: 96x16pixels
MEMORY	• Internal Flash Memory: 192KB • System Memory: 132KB SRAM • Expansion: None



MCU vs SoC

RISC-V MCUs:

- 32 or 64 bit
- RTOS
- Toothbrushes, clocks
- Low power IoT devices

- Can't run Dart

RISC-V SoCs:

- 64 bit
- Linux
- Phones, tablets, set top boxes
- High power IoT devices

- Can run Dart



RISE worked



Goal: Accelerate open source SW for RISC-V architecture

How: Align on highest priorities & avoid (accidental) duplication of work



riscv64 support is now present in major distros :)

- Debian 13 (Trixie)
- Fedora 42
- *RHEL 10 (developer preview)*
- Ubuntu 24.04

But...

Android NDK is now years late

- Originally promised for 2024
- ABI still hasn't been frozen
 - Alignment with standard ISA profiles such as RVA23
- riscv64 generic kernel images removed from Android Common Kernel (ACK)

Is there still hope for CHERI?



RISC-V in the cloud

Elastic Metal RV1

The world's first RISC-V servers available in the cloud.

Taste the new open processor architecture now.

Will you take the risk?

Order Now →

Read the press release →



<https://labs.scaleway.com/en/em-rv1/>

But...



Ghost Write

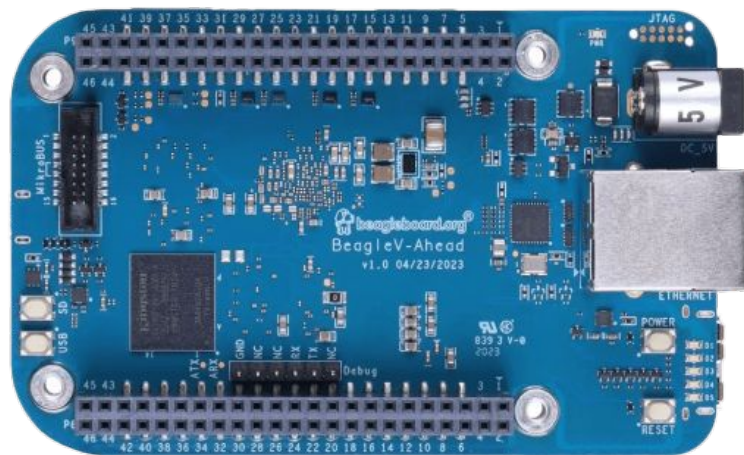
<https://www.infoq.com/news/2024/08/GhostWrite/>

Hardware vapourware

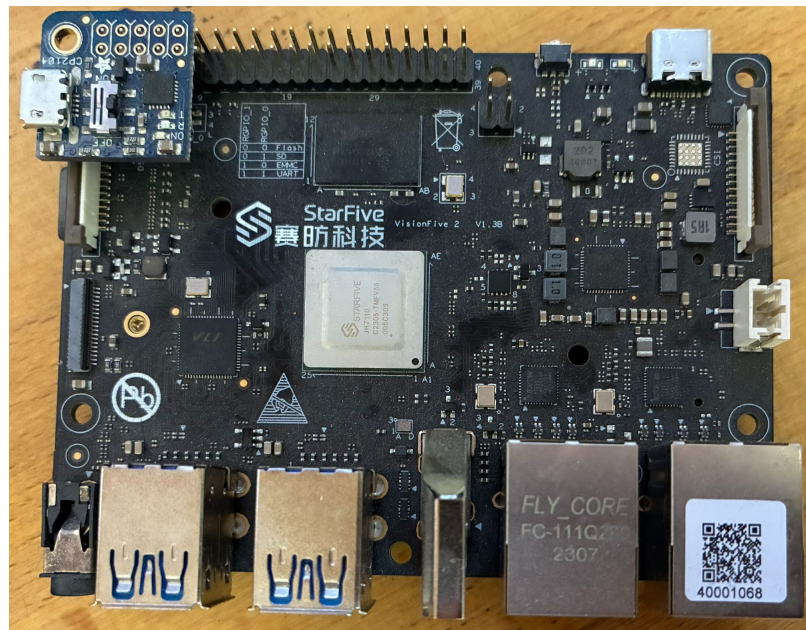
The boards that never came

- Milk-V Oasis
- SiFive HiFive Pro P550 (Intel "Horse Creek" SoC)
- ...

Dev board drought



2023



2023

The stalking horse conjecture



The China Factor

RISC-V business: Tech foundation moving to Switzerland because of geopolitical concerns

Unanimous decision of board to up sticks from Delaware

 Tim Anderson

Tue 26 Nov 2019 // 11:35 UTC

The RISC-V Foundation, which directs the development of an open-source instruction set architecture for CPUs, will incorporate in Switzerland. Currently it is a non-stock corporation in Delaware, USA.

https://www.theregister.com/2019/11/26/riscv_foundation_switzerland/

Huawei running out of smartphone CPUs as US sanctions begin to bite

Chinese giant suggests current Kirin SoC could be last of its kind

 [Robbie Harb](#)

Mon 10 Aug 2020 // 05:02 UTC

Huawei will halt production of its flagship Kirin chipsets later this year, due to US sanctions on Chinese companies.

Production of the Chinese firm's Kirin and AI-powered chips, both of which are used to power its handsets, will be forced to stop on September 15 as a result of US sanctions on its suppliers, said Richard Yu, president of the company's consumer unit, at an event on Friday.

https://www.theregister.com/2020/08/10/huawei_halts_chip_production_as/



New shoots...

Accelerating Software Optimization, Porting and Validation for Datacenters

The SiFive BigSky SF-2U870 datacenter development platform delivers on the promise of the RISC-V open standard. It is a ready-to-go, out-of-the-box, fully integrated commercial RISC-V datacenter development platform built in a rackable 2U server form factor.

- ✓ SiFive BigSky P870-D Compute Complex
- ✓ Ubuntu 26.04 LTS
- ✓ 32x P870-D cores operating at 2.0 GHz (TSMC N6)
- ✓ 256GB DDR5-5600
- ✓ 4 PCIe Gen 5 x16 (64 lanes) and PCIe Gen 3 x4
- ✓ Support for x16 double-wide GPU
- ✓ 2x 7.68TB U.2 NVMe SSD
- ✓ 10/25Gb OCP 3.0 NIC



<https://www.sifive.com/development-platforms/sifive-bigsby-sf-2u870-datacenter>

Review

- RISC-V looked inevitable, but there's been a lull
- Mainstream distro support arrived on time
- Android seems stuck
- China pivot hasn't yet had a noticeable impact
- Testing instances are available but weedy
 - But maybe it's just that hardware takes time

Thanks for your time

@cpswan

Questions?